

Python Project Collection

This document showcases beginner-to-intermediate Python projects built as part of a learning journey. Most of these projects were guided by Tech With Tim's tutorials, with the exception of the Calculator UI project, which was based on Devression's design.

Transaction Terminal

Reference: Tech With Tim

A simple command-line banking system that lets users:

- Check balance
- Deposit funds
- Withdraw funds

This project demonstrates the use of functions, loops, and basic user input handling.

WPM Typing Test

Reference: Tech With Tim

A terminal-based typing speed test built using the curses library.

Features:

- Randomly loads a line from a file
- Calculates real-time Words Per Minute (WPM)

This project introduces working with the curses module, file I/O, and live input handling.

Password Manager

Reference: Tech With Tim

This project includes two versions:

- Basic version: Stores account names and passwords in plain text
- Advanced version: Uses the cryptography library to encrypt and decrypt passwords

It covers file management, basic encryption, and simple user authentication.

Slot Machine Game

Reference: Tech With Tim

A command-line slot machine simulator that includes:

- Randomized spinning symbols
- Multi-line betting system
- Balance tracking and win/loss logic

This reinforces logic building, randomness, and loop-based control.

Currency Converter

Reference: Tech With Tim

Fetches real-time currency exchange rates from currconv.com API.

Supports:

- Listing available currencies
- Checking exchange rates
- Converting currency amounts

The project teaches how to use APIs, handle JSON data, and perform real-time calculations.

Calculator UI

Reference: Devression

A functional calculator with a graphical user interface using Tkinter.

Features:

- Basic operations (+, -, ×, ÷)
- Percentage, sign change, and clear functionality
- Expression evaluation with Python's built-in logic

This combines GUI layout with mathematical logic.

REQUIREMENTS

- Python 3.x
- cryptography (for encrypted password manager)
- requests (for currency API)
- tkinter (comes pre-installed with most Python distributions)

To install required packages:

```
pip install cryptography requests
```

ACKNOWLEDGEMENTS

Tech With Tim (YouTube): <https://www.youtube.com/c/TechWithTim>

Devression (YouTube): <https://www.youtube.com/@Devression>

NOTES

Each project is self-contained in its own .py file.

To run a project:

1. Open a terminal
2. Navigate to the file's directory
3. Use `python filename.py` to run